

Newton's Laws of Motion

Subject: Physics

Grade: Grade 10

Explore Isaac Newton's three laws and how they describe motion in our everyday world.

First Law - Inertia

An object stays at rest or continues in motion unless acted on by an outside force. This is why a book stays still until you push it.

Second Law - Force and Acceleration

Force equals mass times acceleration ($F = m \times a$). A stronger force or lighter mass causes more acceleration.

Third Law - Action and Reaction

For every action, there is an equal and opposite reaction. When you jump, your legs push the ground and the ground pushes you up.